

Lab Evaluation Question

Synthetic Image Generation and Model Comparison using VAE and GAN

Problem Statement:

You are provided with a multi-class mango leaf disease image dataset. Your task is to apply two generative models:

- Variational Autoencoder (VAE)
- Generative Adversarial Network (GAN)

to generate synthetic images and compare the performance of the two models in terms of image quality and diversity.

Tasks to Perform:

1. Preprocess the dataset (resize, normalize, and organize images class-wise).
2. Implement and train a Variational Autoencoder (VAE) on the dataset.
3. Implement and train a GAN (DCGAN or StyleGAN2-ADA) on the same dataset.
4. Generate at least 100 synthetic images from each model.
5. Compare the two models using:
 - Visual comparison (sample image grids)
 - Quantitative metrics, such as:
 - Structural Similarity Index (SSIM)
 - Fréchet Inception Distance (FID)
6. Submit a short report highlighting:
 - Model architecture summaries
 - Training process and challenges
 - Visual samples and metric scores
 - Which model performed better and why?

Deliverables:

- VAE and GAN implementation code
- 100+ synthetic images per model

- Evaluation plots or metric scores
- Final report (2-4 pages or Jupyter notebook)

Evaluation Rubric (Out of 20 Marks):

- Dataset preprocessing: 2 marks
- VAE implementation and results: 4 marks
- GAN implementation and results: 4 marks
- Model comparison (visual + metrics): 6 marks
- Report quality and insight: 4 marks